

Asian College of Higher Studies

Ekantakuna, Lalitpur



Lab Report

Department : Computer Science and Information Technology
Course Title : Software Project Management
Course No : CSC 425
Experiment No : 04
Experiment Name : Cost Benefit Analysis (CBA)

Submitted By

Name: Sulav Adhikari
Id: 23081001
Level: Bachelor
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Submitted To

Name of the Teacher:
Mr Subhash Paudel
Designation:
Lecturer
Signature:

Lab 4: Cost Benefit Analysis (CBA) for Software Project Investment Decision

①) We need to develop an online student Admission Management system for a University which includes student registration, document upload, online payment, merit list generation, & administrative reporting. Some activities can be executed in parallel to reduce project duration.

Now find the following information to perform a Cost-Benefit Analysis:

- 1) Calculate the Net Present Value (NPV) of the project.
- 2) Calculate the Internal Rate of Return (IRR).
- 3) Calculate the payback period.
- 4) Compute the Benefit-Cost Ratio (BCR).
- 5) Draw a simple cash flow diagram.
- 6) Interpret whether the project is financially feasible.

Assume the following data:

- Software Development Cost = NPR 800,000
- Hardware & Infrastructure Cost = NPR 300,000
- Training Cost = NPR 200,000
- Annual Maintenance Cost = NPR 100,000
- Discount Rate = 10%.

The expected annual benefit as:

Year	Expected Benefit (NPR)
Year 1	500,000
Year 2	600,000
Year 3	750,000
Year 4	850,000
Year 5	900,000

Solution :

Given Assumptions;

Amount (NPR)	Cost Component
800,000	Software Development Cost
300,000	Hardware & Infrastructure Cost
100,000	Training Cost
200,000	Setup Cost
1,400,000	Total initial investment (Co)

Other Assumptions:

Item	Value
Annual Maintenance	NPR = 100,000 per Year
Discount Rate (%)	10% = 0.10
Project Life	5 Years

Expected Annual Benefits & Net cash Flow:

• $NCF = \text{Annual Benefits} - \text{Annual Maintenance cost}$

Year	Benefit (NPR)	Maintenance (NPR)	Net cash Flow (NPR)
Year 1	500,000	100,000	$500,000 - 100,000 = 400,000$
Year 2	600,000	100,000	$600,000 - 100,000 = 500,000$
Year 3	750,000	100,000	$750,000 - 100,000 = 650,000$
Year 4	850,000	100,000	$850,000 - 100,000 = 750,000$
Year 5	900,000	100,000	$900,000 - 100,000 = 800,000$

1. Net Present Value (NPV)

$$\text{Formula: } NPV = -C_0 + \frac{NCF(1)}{(1+r)^1} + \frac{NCF(2)}{(1+r)^2} + \dots + \frac{NCF(4)}{(1+r)^4} + \frac{NCF(5)}{(1+r)^5}$$

Where, C_0 = Initial Investment, r = Discount rate

$NCF(t)$ = Net Cash Flow in Year t

$$\begin{aligned} NPV &= -1,400,000 + \frac{400,000}{(1.10)^1} + \frac{500,000}{(1.10)^2} + \frac{650,000}{(1.10)^3} + \\ &\quad \frac{750,000}{(1.10)^4} + \frac{800,000}{(1.10)^5} \\ &= -1,400,000 + 363,636 + 413,223 + 488,355 + \\ &\quad 512,260 + 496,737 \\ &= -1,400,000 + 2,274,211 \\ &= 874,211 \end{aligned}$$

Detailed PV factor calculation for each Year:

Year	NCF (NPR)	PV factor = $1/(1.10)^t$	Calculation	PV of NCF (NPR)
1	400,000	$1/(1.10)^1 = 0.9091$	$400,000 \times 0.9091$	363,636
2	500,000	$1/(1.10)^2 = 0.8264$	$500,000 \times 0.8264$	413,223
3	650,000	$1/(1.10)^3 = 0.7513$	$650,000 \times 0.7513$	488,355
4	750,000	$1/(1.10)^4 = 0.6830$	$750,000 \times 0.6830$	512,260
5	800,000	$1/(1.10)^5 = 0.6209$	$800,000 \times 0.6209$	496,737
Total	3,100,000	—	—	2,274,211

$$\begin{aligned} \therefore NPV &= \text{Total PV of Net Cash Flow (NCF)} - \text{Initial Investment } (C_0) \\ &= 2,274,211 - 1,400,000 \\ &= \text{NPR } 874,211 \end{aligned}$$

Hence, the result is positive & the project is financially worthwhile.

Since $NPV > 0$, the project earns more than the 10% discount rate over 5 years.

2) Internal Rate of Return (IRR):

Formula: IRR = the discount rate at which $NPV = 0$.

Set $NPV = 0$ & solve for r .

$$0 = -C_0 + \frac{NCF(1)}{(1+r)^1} + \frac{NCF(2)}{(1+r)^2} + \frac{NCF(3)}{(1+r)^3} + \frac{NCF(4)}{(1+r)^4} + \frac{NCF(5)}{(1+r)^5}$$

Method: use trial & error, then apply linear interpolation to find exact IRR.

Step 1 - trial & error (substitute different rates until NPV changes sign).

Trial rate (r)	NPV Calculation	NPV result	Decision
$r = 10\%$	$-1,400,000 + 2,274,211$	$= +874,211$	Pos - increase r
$r = 20\%$	$-1,400,000 + 1,868,091$	$= +468,091$	Pos - increase r
$r = 25\%$	$-1,400,000 + 1,634,212$	$= +234,212$	Pos - increase r
$r = 29\%$	$-1,400,000 + 1,414,312$	$= +14,312$	Very close to zero
$r = 30\%$	$-1,400,000 + 1,385,215$	$= -14,785$	Neg, $\rightarrow$ IRR between 29% & 30%

Step 2: Linear Interpolation Formula

$$IRR = r(L) + \left[\frac{NPV(L)}{NPV(L) + |NPV(H)|} \right] \times (r(H) - r(L))$$

where $r(L)$ = lower rate, $r(H)$ = higher rate $NPV(L)$ = NPV at lower rate $NPV(H)$ = NPV at higher rate.

$$\begin{aligned} IRR &= 29\% + \left[\frac{14,312}{14,312 + 14,785} \right] \times (30 - 29\%) \\ &= 29\% + \left[\frac{14,312}{29,097} \right] \times 1\% \\ &= 29\% + 0.4918 \times 1\% \\ &= 29.26\% \end{aligned}$$

Result $\Rightarrow$ IRR = 29.26% > Discount rate (10%)

Since, IRR > Discount rate, the project earns well above the required rate of return.

3) Payback Period:

Formula:

$$\text{Payback Period} = A + (B/C)$$

where, A = Year just before full recovery (last year cumulative NCF < 0)

B = Amount still to recover at start of recovery year = 60 - Cumulative NCF at end of year A

C = NCF of the recovery year.

Step 1: Cumulative NCF Table:

Year	NCF (NPR)	Cumulative NCF (NPR)	Amount still to recover (NPR)
1	400,000	400,000	$1,400,000 - 400,000 = 1,000,000$
2	500,000	900,000	$1,400,000 - 900,000 = 500,000$
3	650,000	1,550,000	Fully recovered in 3 years
4	750,000	2,300,000	-
5	800,000	3,100,000	-

Step 2: Identify A, B, C & calculate exact payback period.

A = 2 (Cumulative after 2 year = 900,000 < 1,400,000)

B = 1,400,000 - 900,000 = 500,000 (Amount still unrecovered)

C = 650,000 (NCF of year 3, the recovery year)

$$\text{Payback Period} = A + (B/C)$$

$$= 2 + (500,000 / 650,000)$$

$$= 2.77 \text{ years}$$

Results: Payback period = 2.77 years (about 2 year 9 months)

Since Payback period (2.77 years) < project life (5 yrs), Investment is recovered quickly

4) Benefit - Cost Ratio (BCR)

Formula:

$BCR = \text{PV of total benefits} / \text{PV of total costs}$
 Both benefits & costs are discounted to present value using $r = 10\%$.

If $BCR > 1 \Rightarrow$ Project benefits outweigh costs $\Rightarrow$ project is financially beneficial.

Year	Benefit (NPR)	PV Factor = $1/(1.1)^t$	PV of Benefit (NPR) Benefit $\times$ PV Factor
1	500,000	0.9091	454,545
2	600,000	0.8264	500 495,868
3	750,000	0.7513	563,484
4	850,000	0.6836	580,754
5	900,000	0.6209	558,639
Total	3,600,000	—	2,653,290

Step 2: PV total costs

Cost Item	Calculation	PV (NPR)
Initial investment (Year 0)	No discounting - Paid at start	1,400,000
Maintenance Year 1	$100,000 \times 0.9091$	90,909
Maintenance Year 2	$100,000 \times 0.8264$	82,645
Maintenance Year 3	$100,000 \times 0.7513$	75,131
Maintenance Year 4	$100,000 \times 0.6836$	68,361
Maintenance Year 5	$100,000 \times 0.6209$	62,092
Total PV of Costs	$1,400,000 + 378,079$	1,778,079

Step 3: Apply BCR formula:

$$\begin{aligned}
 BCR &= \text{PV of benefits} / \text{PV of costs} \\
 &= 2,653,290 / 1,778,079 \\
 &= 1.49
 \end{aligned}$$

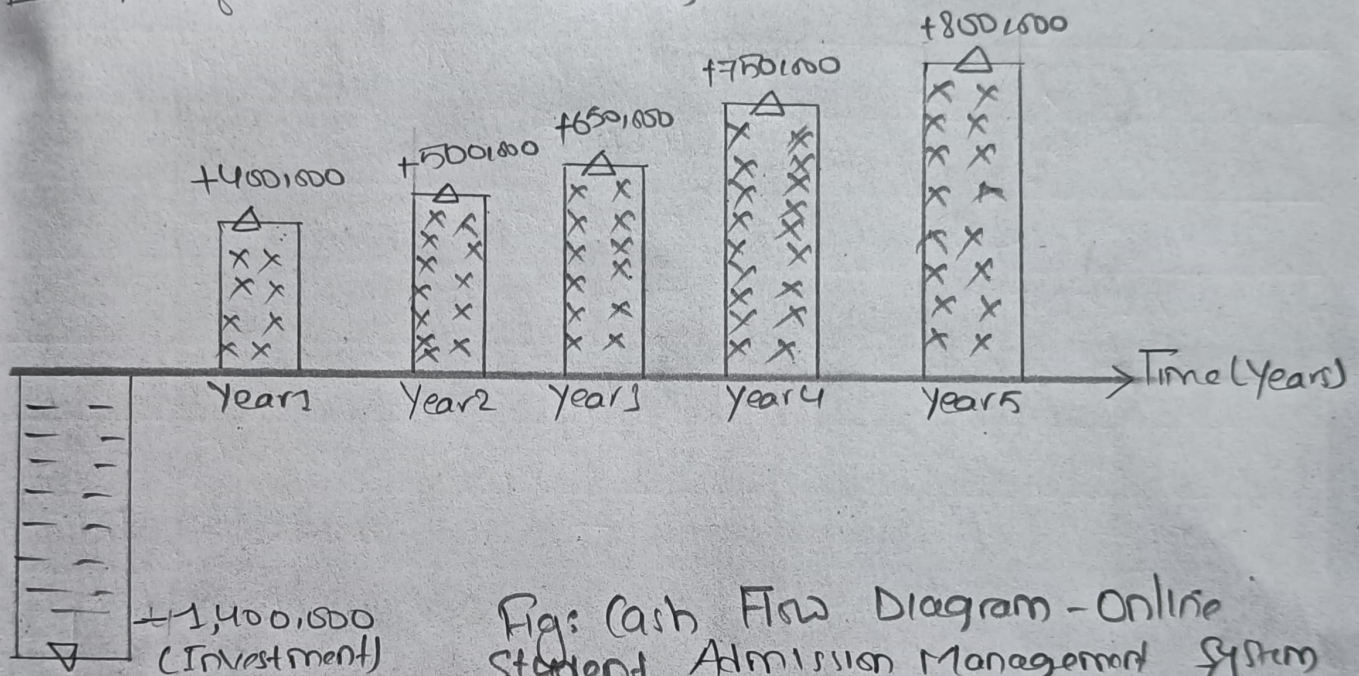
Since $BCR > 1$, every NPR 1 invested return NPR 1.49 therefore project is financially beneficial.

5) Cash Flow Diagram

→ Downward arrow = outflow (Cost/Investment)
 → Upward arrow = inflow (Net benefit).

□ = Outflow (Cost)

xx = Net flow (Benefit - maintenance)



Figs Cash Flow Diagram - Online Student Admission Management System

6) Financial Feasibility Interpretation

Metric	Calculated Value	Criterion	Verdict
NPV	NPR 874,211	> 0 (positive)	Feasible
IRR	29.26%	$>$ Discount rate (10%)	Feasible
Payback Period	2.77 Years	$<$ Project life (5 yrs)	Feasible
BCR	1.49	> 1.0	Feasible

Conclusion: The project is financially feasible therefore recommended.

All four financial indicators confirm this is a sound investment.

a) NPV = NPR 874,211 (positive) → project creates value above cost capital.

b) IRR = 29.26% → the return is nearly 3x the required 10% hurdle rate.

c) Payback = 2.77 year → investment recovered well within the 5 year Project life.

d) BCR = 1.49 → every NPR 1 invested returns NPR 1.49 in discounted benefits.